Seok-Jin Kang

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Bibliography

Seok-Jin Kang, Ph.D. is an **immunologist** and **computational biologist** whose academic and research trajectory bridges traditional immunology and modern **data-driven biology**. During his M.S. and Ph.D. training in the **Immune Modulation Laboratory** *(PI: Prof. Taehoon Chun)* at **Korea University**, he investigated **immune regulation**, **macrophage polarization**, and the **molecular mechanisms** of **host–pathogen interactions** using both **in vitro** and **in vivo** models.

Recognizing that future breakthroughs would arise from integrating **experimental** and **computational** disciplines, Dr. Kang transitioned from wet-lab immunology to **data-driven research**. His expertise now spans **computational biology**, integrating **machine learning**, **deep learning**, and **quantum-inspired approaches** with **multi-omics** datasets. Recent work focuses on the **modeling and identification of intrinsically disordered regions (IDRs)** using **quantum neural networks**, as well as applying **computer-aided drug design (CADD)** to uncover the **biophysical mechanisms** underlying **protein function and regulation**.

In the short term, his research aims to advance **precision medicine** through the development of **multi-omics-based computational models** capable of predicting **immune responses**, guiding **therapeutic design**, and enabling **personalized intervention strategies**. Over the long term, he aspires to establish himself as an **independent faculty member** and contribute to extending **human healthspan and wellbeing** through integrative, **AI-driven biomedical research** that unites **immunology**, **genomics**, and **computational modeling**.

Research Interests

Precision Medicine, CAR-T Cell Therapy, Multi-omics Integration, Computer-aided Drug Design (CADD), Quantum Computing

Education

Korea University — **Mar 2025 – Present**
Postdoctoral Fellow, Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)

Korea University — **Mar 2018 – Feb 2025**
M.S. & Ph.D. in Biotechnology
Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)

Korea University — **Mar 2014 – Feb 2018**
B.S. in Biotechnology

Publications

** indicates first author; † indicates principal investigator.*

Featured Publications

5. ***SJ Kang****, H Shin†, Biophysical mechanisms of spider-silk constituting element–induced stick-slip behavior and hydrogen bond regeneration for high toughness in silk fibers, *International Journal of Biological Macromolecules*, 147027, 2025

4. ***SJ Kang****, H Shin†, Amino acid sequence-based IDR classification using ensemble machine learning and quantum neural networks, *Computational Biology and Chemistry*, 108480, 2025

3. ***SJ Kang****, J Yang, NY Lee, CH Lee, IB Park, SW Park, HJ Lee, HW Park, HS Yun, T Chun†, Monitoring cellular immune responses after consumption of selected probiotics in immunocompromised mice, *Food Science of Animal Resources* 42 (5), 903, 2022

2. ***SJ Kang****, IB Park, T Chun†, Open reading frame 5 protein of porcine circovirus type 2 induces RNF128 (GRAIL) which inhibits mRNA transcription of interferon-β in porcine epithelial cells, *Research in Veterinary Science* 140, 79–82, 2021

1. ***SJ Kang****, T Chun†, Structural heterogeneity of the mammalian polycomb repressor complex in immune regulation, *Experimental & Molecular Medicine* 52 (7), 1004–1015, 2020

Under Review

- ***SJ Kang****, H Shin†, Quantum-Enhanced Transfer Learning for IDR Binding Partner Prediction, *IEEE Transactions on Computational Biology and Bioinformatics*, under review (Round 1), 2025

Published Articles

- JH Yoon*, GB Yeon*, H Lee, H An, J Oh, ***SJ Kang***, IB Park, SA Lim, SS Hwang, DS Kim, JH Kim, T Chun†, YX Fu, J Bae, Tumor-targeted delivery of CXCL10 by mesenchymal stromal cells potentiates adoptive T cell therapy to treat solid tumors, *Biomedicine & Pharmacotherapy* 192, 118579, 2025
- CH Lee*, HJ Lee, SW Park, J Shin, ***SJ Kang***, IB Park, HK Kim, T Chun†, Mutational analysis of pig tissue factor pathway inhibitor α to increase anti-coagulation activity in pig-to-human xenotransplantation, *Biotechnology Letters* 46 (4), 521–530, 2024
- SW Park*, IB Park*, ***SJ Kang***, J Bae, T Chun†, Interaction between host cell proteins and open reading frames of porcine circovirus type 2, *Journal of Animal Science and Technology* 65 (4), 698, 2023
- SP Choi*, SW Park, ***SJ Kang***, SK Lim, MS Kwon, HJ Choi, T Chun†, Monitoring mRNA expression patterns in macrophages in response to two different strains of probiotics, *Food Science of Animal Resources* 43 (4), 703, 2023
- CY Choi*, CH Lee, J Yang, ***SJ Kang***, IB Park, SW Park, NY Lee, HB Hwang, HS Yun, T Chun†, Efficacies of potential probiotic candidates isolated from traditional fermented Korean foods in stimulating immunoglobulin A secretion, *Food Science of Animal Resources* 43 (2), 346, 2023
- SP Choi*, YC Choi, J Yang, CY Choi, CH Lee, ***SJ Kang***, IB Park, T Chun†, Monitoring mRNA transcription of genes involved in early pregnancy from endometrium and peripheral blood mononuclear cells of pregnant pigs with different parity, *Reproduction in Domestic Animals* 53 (6), 1594–1599, 2018
- CY Choi*, YC Choi, IB Park, CH Lee, ***SJ Kang***, T Chun†, The ORF5 protein of porcine circovirus type 2 enhances viral replication by dampening type I interferon expression in porcine epithelial cells, *Veterinary Microbiology* 226, 50–58, 2018

Patents

Registration

- T Chun, J Yang, CH Lee, ***SJ Kang***, IB Park, Chimeric antigen receptor specifically binding to CD38 and use thereof, *Korean Intellectual Property Office*, 10-2021-0025895, October 2021
- T Chun, J Yang, CH Lee, ***SJ Kang***, IB Park, Chimeric antigen receptor specifically binding to VLA-4 and use thereof, *Korean Intellectual Property Office*, 10-2021-0025905, March 2021

Pending

- ***SJ Kang***, H Shin, Method and system for classifying intrinsically disordered proteins based on amino acid sequences using quantum neural networks, *Korean Intellectual Property Office*, 10-2025-0078871, June 2025
- T Chun, IB Park, ***SJ Kang***, HJ Lee, Regulation of BCL11A expression by LDB1 transcriptional regulator and uses thereof, *Korean Intellectual Property Office*, 10-2025-0010845, January 2025
- T Chun, IB Park, ***SJ Kang***, HJ Lee, Regulation of PLZF expression by LDB1 transcriptional regulator and uses thereof, *Korean Intellectual Property Office*, 10-2025-0010844, January 2025
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, ***SJ Kang***, SW Park, HJ Lee, Chimeric antigen receptor specifically binding to CD138 and use thereof, *Korean Intellectual Property Office*, 10-2023-0058232, May 2023
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, ***SJ Kang***, SW Park, HJ Lee, Recombinant protein which recognizes CADM1 and the composition comprising the same for treating cancer, *Korean Intellectual Property Office*, 10-2023-0061710, May 2023

Certifications

- **Big Data Analysis Engineer**, *Korea Data Agency*, 2025, BAE-010002862
- **ADsP (Advanced Data Analytics Semi-Professional)**, *Korea Data Agency*, 2024, ADsP-040004911

Research Projects

13. **Q2431261** *02/2025 – 12/2025*
    Title: Safety evaluation of viral vector-based gene therapy with respect to non-specific genome insertion
    Source: Ministry of Food and Drug Safety
    Role: Research Assistant *(PI: Prof. Taehoon Chun)*
    - **Lentiviral/AAV vector engineering** and **murine model** administration
    - **WGS/NGS library** construction and sequencing for integration profiling
    - **Statistical correlation analysis** of insertion sites by dose, route, and vector type

12. **Q2515581** *05/2025 – 12/2025*
    Title: Optimization of HLA gene editing for the development of allogeneic stem cell therapy
    Source: Korea National Institute of Health
    Role: Research Assistant *(PI: Prof. Dongho Geum)*
    - **HLA-edited iPSC generation** for **hypoimmunogenic cell line** establishment using **CRISPR-Cas9**
    - **Directed differentiation** toward **immune-evasive** cardiomyocytes and T cell
    - **In vivo validation** of immune compatibility and engraftment efficiency in **humanized murine models**

11. **R2222761** *09/2022 – 02/2025*
    Title: Regulation of immune cell differentiation by chromatin 3D structural changes
    Source: National Research Foundation of Korea
    Role: Research Assistant *(PI: Prof. Taehoon Chun)*
    - **Chromatin 3D reorganization analysis** mediated by **LDB1 looper** across immune cell types
    - **Multi-omics integration** (RNA-seq, ATAC-seq, ChIP-seq) for enhancer–promoter interaction mapping
    - **In vivo functional validation** using **floxed (Cre–LoxP) mouse models** to assess LDB1-dependent gene regulation

10. **R2212861** *04/2022 – 12/2023*
    Title: Development of universal SARS-CoV-2 and sarbecovirus vaccine candidate using NDV vector platform
    Source: Korea Health Industry Development Institute
    Role: Research Assistant *(PI: Prof. Kisoon Kim)*
    - **Computational modeling** of viral antigen epitopes using **AlphaFold-based structural biology** to enhance cross-neutralizing immune responses
    - **In vitro and in vivo immunogenicity evaluation** in animal models
    - **Immune response profiling** through **ELISA** and **flow cytometry** analyses

9. **R2131151** *02/2022 – 12/2025*
   Title: Development of cellular immune evaluation technology for viral vector-based genetic vaccines
   Source: Ministry of Food and Drug Safety
   Role: Research Assistant *(PI: Prof. Taehoon Chun)s*
   - **Selection of vaccine candidates** including **adenovirus-based (COVID-19)** and **vaccinia virus–based (HPV)** platforms
   - **Development of humoral and cellular immune evaluation assays** for viral vector–based vaccines
   - **Safety evaluation technology development** for viral vector vaccines, including cytotoxicity and organ-specific toxicity analyses

8. **R1923081** *09/2019 – 02/2022*
   Title: Mechanistic study of macrophage polarization regulated by chromatin remodeling factor Phc2
   Source: National Research Foundation of Korea
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Transcriptomic profiling** of macrophages under distinct polarization states
   - **RNA-seq–based differential expression** and pathway analysis
   - Identification of **Phc2-dependent regulatory networks** governing macrophage activation

7. **R1727591** *01/2018 – 12/2020*
   Title: Development of DNA markers to enhance resistance against chronic wasting diseases
   Source: Rural Development Administration
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*

- **Sequence analysis** and **SNP genotyping** to identify host resistance markers
- Development of **computational genomics pipelines** using **R and Python**

6. **R1715822** *03/2018 – 06/2022*
   Title: Development of novel immune cell therapy targeting multiple myeloma
   Source: National Research Foundation of Korea
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Design and optimization** of **chimeric antigen receptor (CAR)** constructs through protein engineering and molecular cloning
   - **Target-binding protein selection** based on affinity and specificity
   - **Binding kinetics measurement** (K_d) using **surface plasmon resonance (SPR)** for functional validation

5. **Q2029461** *01/2021 – 06/2023*
   Title: Production and efficacy evaluation of novel recombinant immunotherapeutics
   Source: Immunological Designing Lab Co., Ltd.
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Recombinant therapeutic protein production** through molecular cloning and protein purification
   - **Binding affinity evaluation** using **SPR** and **ELISA** assays

4. **Q2015891** *06/2020 – 02/2021*
   Title: Screening of immune-enhancing probiotics using immunodeficient mouse models
   Source: CJ CheilJedang & BLOSSOM PARK
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Animal experimentation** and **cytokine profiling** via **flow cytometry** and **qPCR**
   - **Statistical analysis** in **R** to evaluate immune enhancement

3. **Q1815701** *06/2018 – 11/2018*
   Title: Determination of dosing period and amount of probiotics for alleviating allergic rhinitis
   Source: CJ CheilJedang
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Evaluation of probiotic-induced modulation** of **Th1/Th2 cytokine ratios** using **ELISA** and gene expression profiling
   - **Statistical modeling** to identify optimal probiotic dosing for immune regulation

2. **PJ013271012018** *03/2018 – 12/2020*
   Title: Development of genetic markers associated to the resistance to PMWS
   Source: Rural Development Administration
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Comparative genomics** and **sequence-based variant analysis** for resistance marker identification
   - **Bioinformatics pipeline implementation** using **Biopython** and **BLAST** to detect resistance-associated mutations

1. **Q1613841** *07/2016 – 04/2017*
   Title: Selection and in vivo efficacy verification of probiotics with enhanced IgA secretion
   Source: CJ CheilJedang
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Measurement of mucosal IgA levels** in mouse models to identify strains with strong immunostimulatory potential
   - **Gene expression validation** using **qPCR** for immune response assessment

Book chapters

International Publications

- ***SJ Kang***, **2026 Science Trends, Chapter: Multi-omics**, *Kindle*, ISBN 9798266588004, September 2025
- ***SJ Kang***, **2025 Science Trends, Chapter: Artificial Intelligence**, *Kindle*, ISBN 979-8307079270, January 2025

Domestic Publications

- ***SJ Kang***, **2026 생명과학트렌드, 챕터: 멀티오믹스(2026 Life Science Trends, Chapter: Multi-omics)**, *Pubple*, ISBN 9788924169034, August 2025
- ***SJ Kang***, **2026 과학트렌드, 챕터: 개인 맞춤형 치료(2026 Science Trends: Personalized Medicine)**, *Pubple*, ISBN 9788924169041, August 2025
- ***SJ Kang***, **2025 과학트렌드, 챕터: 인공지능(2025 Science Trends, Chapter: Artificial Intelligence)**, *Pubple*, ISBN 9788924143621, December 2024